

EXPERIMENT 04

Analyze Email Headers and Detect Email Spoofing Using MHA (Mail Header Analyzer)

OBJECTIVE

To access, extract, and analyze raw email internet headers, trace routing MTA servers via **Received:** fields, verify email authentication mechanisms (SPF, DKIM, DMARC), and evaluate potential email spoofing.

TOOLS / REQUIREMENTS

- **Email Platform / Client:** Gmail Web Interface
- **Analysis Tools:** Text Editor (Notepad) / Visual Email Header Inspection
- **Online Verification Utilities:** IP Geolocation & WHOIS Lookup (WhatIsMyIP / LookIP.net)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

An institutional email communication was analyzed to determine authenticity, trace transmission infrastructure, and identify potential spoofing indicators. The investigator accessed the RFC 822 / MIME header, parsed MTA intermediate routing hops, executed IP lookup on origin servers, and verified cryptographic SPF, DKIM, and DMARC alignment.

EVIDENCE / INPUT

- **Target Evidence File:** GATE 2027 & JAM 2027 Examinations - Reg.eml
- **Sender Address:** "Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
- **Recipient Address:** csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
- **Message-ID:** <CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=c0bjpLXg@mail.gmail.com>
- **Delivery Time:** Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
- **Sending MTA Host IP:** 209.85.220.69 / 209.85.220.41

PROCEDURE

Step 1 — Accessing the Email in Mail Client

The target email ("Fwd: GATE 2027 & JAM 2027 Examinations - Reg") was opened in the Gmail interface to inspect sender headers, delivery details, and message context.

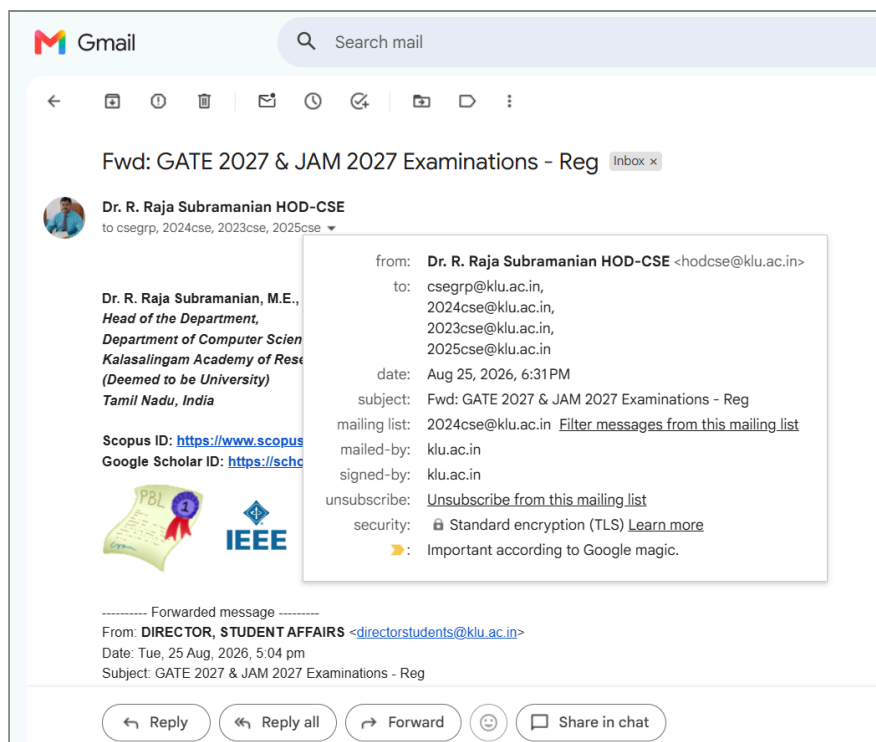


Figure 4.1: Target email accessed in mail client

Step 2 — Viewing Original Message Metadata

The *Show original* option was selected from the message menu, opening the RFC header overview displaying Message ID, creation timestamp, From, and To fields.

Original Message	
Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObj
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025

Figure 4.2: Original message metadata overview

Step 3 — Inspecting Mail Authentication Overview

The authentication overview table was evaluated:

Authentication Check	Evaluation Status	Details
SPF	PASS	Permitted sender IP 209.85.220.69
DKIM	PASS	Domain klu.ac.in signature verified
DMARC	PASS	Policy compliant

Original Message

Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObjpLXg@mail.gmail.com>
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: GATE 2027 & JAM 2027 Examinations - Reg
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

Figure 4.3: Email authentication status table (SPF, DKIM, DMARC)

Step 4 — Extracting Full RFC 822 Raw Headers

The complete original raw MIME header was displayed with *Download Original* and *Copy to clipboard* controls for detailed offline forensic parsing.

Original Message

Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObjpLXg@mail.gmail.com>
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 23 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: GATE 2027 & JAM 2027 Examinations - Reg
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

[Download Original](#)[Copy to clipboard](#)

Delivered-To: 99249041233@klu.ac.in
Received: by 2002:a05:7300:e113:b0:315:b785:cfbb with SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700 (PDT)
X-Forwarded-Encrypted: i=5;
AHgh+RsczupKOTiJLANS719ImE2lRPGAB5LONfLzddL8PzO4UVO1qdYKCuvjHk1dH8bIVgdsAU2AcZCaUxg/w=@klu.ac.in
X-Received: by 2002:a05:6402:505b:b0:698:af31:5a9b with SMTP id 4fb4d7f45d1cf-6a42f1e08c0mr38071921a12.9.1787662895622; Tue, 25 Aug 2026 06:01:35 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787662895; cv=pass; d=google.com; s=arc-20260327; b=M4SEaA7xG61EoGExP2Ho13mduYKPMgBxclMv4-Tu02THFdh51g8akKwuv5B71u0K

Figure 4.4: Full original RFC 822 MIME headers interface

Step 5 — Analyzing Primary Receiving MTA Hop

The raw header was pasted into a text editor (email-header.txt). The top Received: line (Received: by 2002:a05:7300:e113:b0:315:b785:cfbb with SMTP id...) was examined to confirm the final delivering MTA and ARC security seal.

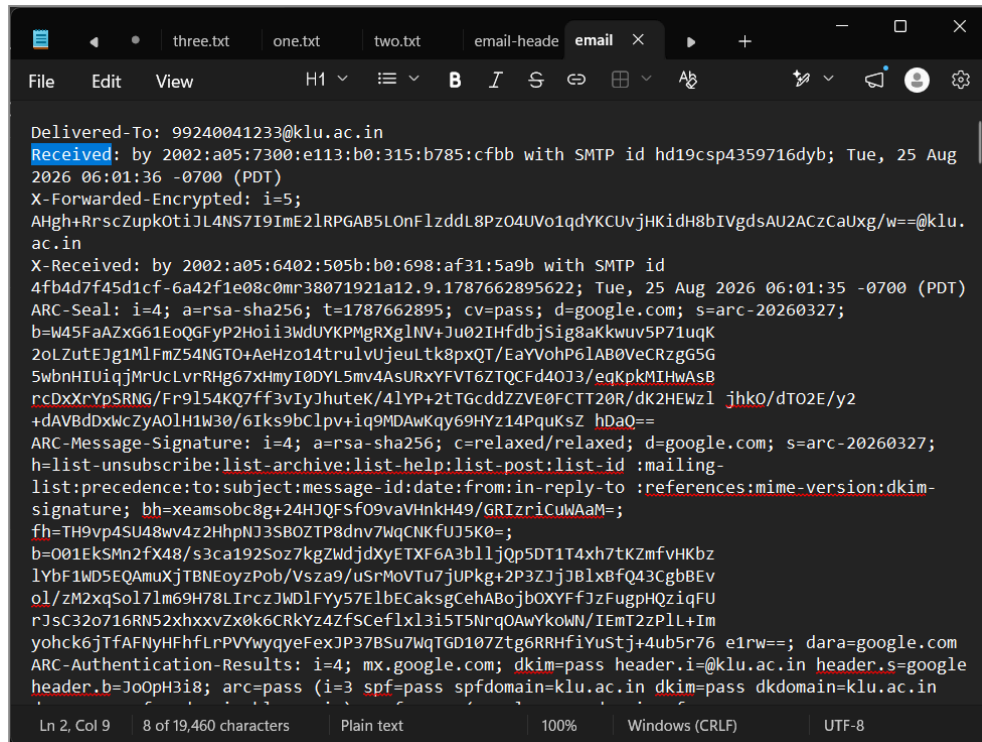


Figure 4.5: Examination of top Received header hop

Step 6 — Reviewing Comprehensive Header Stream

The entire structured header stream was analyzed to verify the chronological relay sequence across intermediate mail transfer agents.

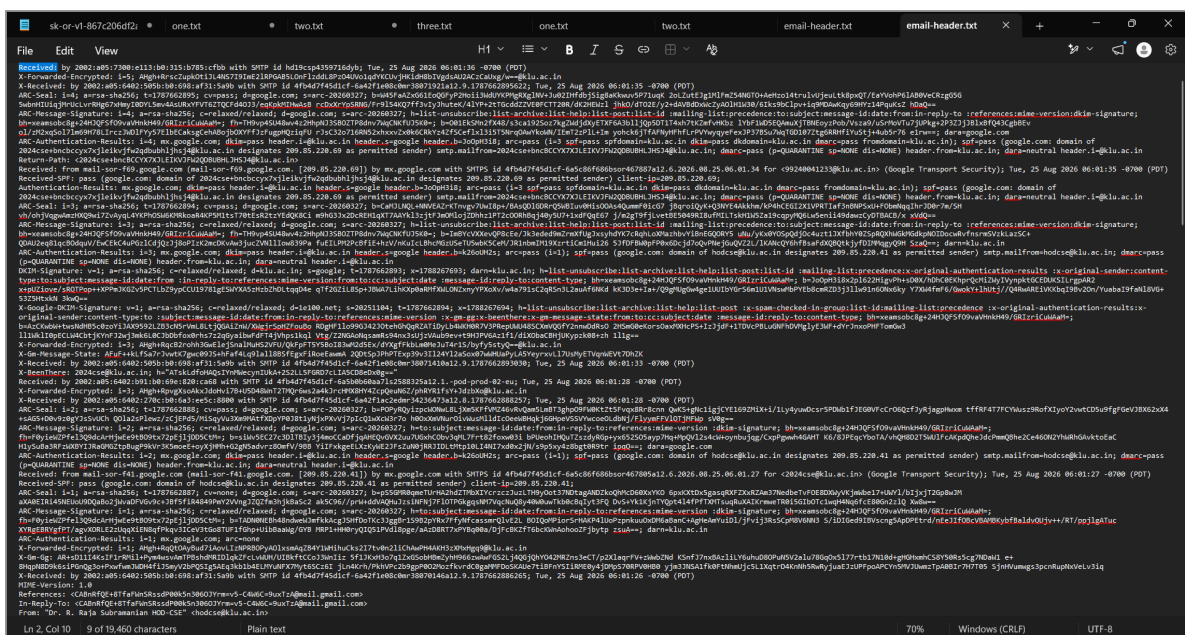


Figure 4.6: Complete raw email headers analysis

Step 7 — Detailed Inspection of Initial SMTP Hop

A focused examination of the initial receiving hop was conducted to verify protocol compliance (SMTP id hd19csp4359716dyb; Tue, 25 Aug 2026 06:01:36 -0700).

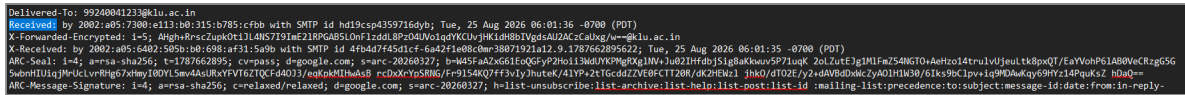


Figure 4.7: Zoomed inspection of initial SMTP receiving hop

Step 8 — Identifying Key Header Fields

Key forensic metadata fields were cataloged from the header:

- From: "Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
- Subject: Fwd: GATE 2027 & JAM 2027 Examinations - Reg
- Date: Tue, 25 Aug 2026 18:31:12 +0530
- Message-ID: <CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=c0bjpLXg@mail.gmail.com>

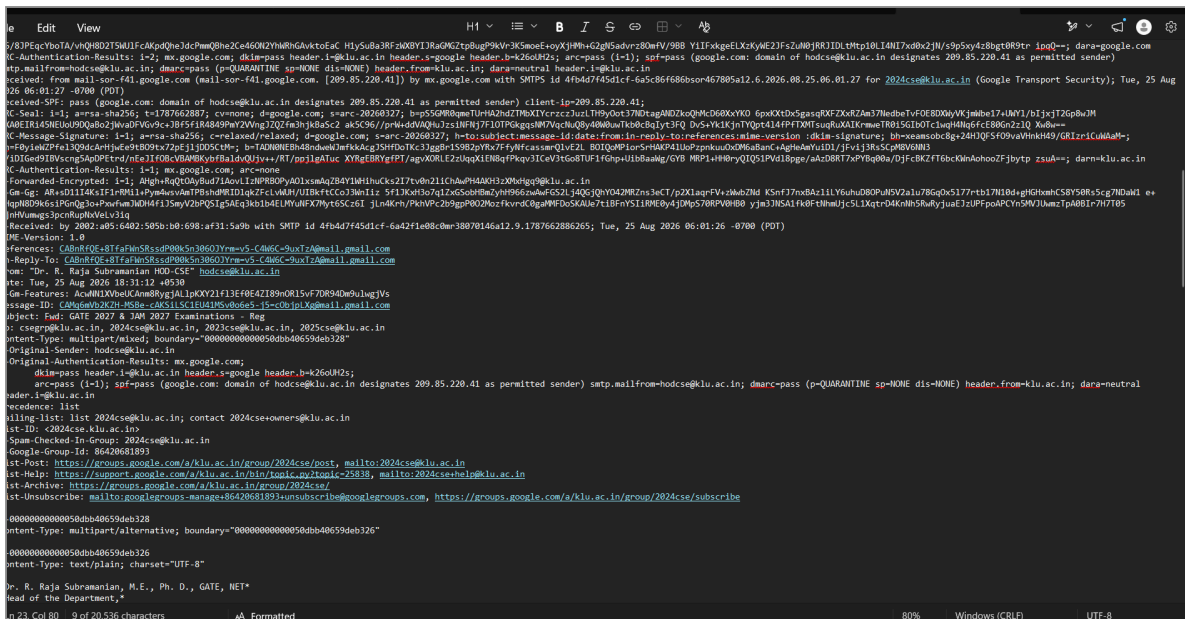


Figure 4.8: Identification of key RFC header fields

Step 9 — Geolocation and WHOIS IP Lookup of Sending Server

The sending relay IP address 209.85.220.69 was queried through an online IP lookup service, confirming:

- **Owner / ISP:** Google LLC
- **Autonomous System:** AS15169
- **CIDR Block:** 209.85.208.0/20
- **Location:** Utica, New York, United States

The IP address 209.85.220.69 is owned by Google LLC and is located in Utica, New York, United States.

Ownership and Network Details

- **Owner/ISP:** Google LLC
- **ASN (Autonomous System Number):** AS15169
- **CIDR Block:** 209.85.208.0/20 (some sources list 209.85.192.0/18)
- **IP Type:** Public IPv4
- **Hostname:** Associated with Google infrastructure, no specific websites detected running directly on this IP [LookIP.net +3](#).

Geolocation

- **City:** Utica
- **State:** New York
- **Country:** United States
- **Time Zone:** Eastern Time Zone (Olson TZ database)
- **Accuracy:** Approximate; IP geolocation is based on ISP registration and routing information, not the exact physical location of a device [WhatIsMyIP.com +1](#).

Figure 4.9: IP lookup and geolocation analysis for sending MTA

Step 10 — Evaluating Sender Policy Framework (SPF) Status

The SPF authentication record in the header was verified: `spf=pass` (google.com: domain of `hodcse@klu.ac.in` designates 209.85.220.41 as permitted sender).

```
Content-Type: multipart/mixed; boundary="00000000000050dbb40659deb328"
-Original-Sender: hodcse@klu.ac.in
-Original-Authentication-Results: mx.google.com;
    dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s;
    arc=pass (i=1); spf=pass (google.com: domain of hodcse@klu.ac.in designates 209.85.220.41 as permitted sender);
    p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in; dara=neutral header.i=@klu.ac.in
Precedence: list
Replying-list: list 2024cse@klu.ac.in; contact 2024cse+owners@klu.ac.in
List-ID: <2024cse.klu.ac.in>
```

Figure 4.10: Sender Policy Framework (SPF) verification in raw header

Step 11 — Verifying DomainKeys Identified Mail (DKIM) Signature

The DKIM cryptographic signature header was evaluated: DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; confirming signature validity and body hash integrity.

```
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s; arc=pass (i=1); spf=pass (google.co
hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.fro
dana=neutral header.i=@klu.ac.in
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1787662893; x=1788267693; darn=klu.ac.in; h=list-unsubscribe:list-arc
help:list-post:list-id :mailing-list:precedence:x-original-authentication-results :x-original-sender:content-type:to:subject:message-id:date:from :i
to:references:mime-version:from:to:cc:subject:date :message-id:reply-to:content-type; bh=xeamsobc8g+24HJQFSf09vaVHnkH49/GRIzriCuWAAm=;
b=JoOpH3i8x2p1622HigVhS+D0X/hDhC0EKhrQcMiZWyIVynpktGCEUDUKSILrnpAR2 x+puZiove/sRQTPop++XPPmJKGZv5PCTLbZ9ypCCU19781gESWYXASzHbZhdLttqD4e
```

Figure 4.11: DKIM cryptographic signature header inspection

Step 12 — Examining Message-ID and ARC Authentication Seal

The ARC (Authenticated Received Chain) signature and message-id header references were analyzed to confirm end-to-end chain verification across forwarding hops.

```
ARC-Seal: i=3; a=rsa-sha256; t=1787662894; cv=pass; d=google.com; s=arc-20260327; b=CaMjLNQL+NNVEAZrKtNvgv7LwI8p+/BAsQD1GDRnQ5WBI0Hs00As4QumF8icG7
jBqro3QyK+Q3NYE4Akkhm/kP4hCEGI2X1VPR1Iaf3nBNPSXuF0bmIqg1nr3D0r7m/SH_vh/oh3VqguAmzhXQ9wi7ZvAyl4YKPhOSN6KMRkoaR4KPSM1ts170tEsR2tzYEaQK8Ci
9hG3Jx2DcREH1qXT7AAVkl3zjtfJmOw1oj2DhHz1PT2c0ORhBqj40ySUT+1xdFQqE67 j/m2g19fjLvetBE5049RI8uFmILTskH1W5Za19capyWQ6Lw5enii49dazwCyDTBACB/x xVdQ==
ARC-Message-Signature: i=3; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-
list:precedence:to:subject: message-id: date: from: in-reply-to : references: mime-version: dkim-signature; bh=xeamsobc8g+24HJQFSf09vaVHnkH49/GRIzriCuWAAm=;
fh=TH9vp4S4M8w4z2HhpNj3S802TP8dnv7nqCNKFUJ5K0; b=ImBYcVXXevQP8cEe/Jk3aded9mZrmXfUg1xshydYK7cRqHLoMazhbvYiBnEGQORY5
uNu/yKx0Y0S0pQdj0c4uzt1JXfbhY8ZSpRQXhMgKgdKpN0IDocwRvfnsrSVzklaz5C+ Q0AU2eq81gcB0dquV/EwCEkC4uPGz1CdJQzj38oPIzK2ncDKvAw3jucZVN1Iow839Pa
FuEILPM2PcBfiE+hzV/nKuIClBhcMgzUSaTUSwbK5CeM/JR1nbmIM19XzntiCm1Hui26 5JFDFBw0pFP0x6Dcdj7oQvPWejguQVZ2L/JKANCQY6hf8sAfDx080tkjyfdIWWagYQ9H Sza0==; darn=klu.ac.in
ARC-Authentication-Results: i=3; mx.google.com; dkim=pass header.i=@klu.ac.in header.s=google header.b=k26oUH2s; arc=pass (i=1); spf=pass (google.com; domain of
hodcse@klu.ac.in designates 209.85.220.41 as permitted sender) smtp.mailfrom=hodcse@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header.from=klu.ac.in;
```

Figure 4.12: Message-ID and ARC authentication seal verification

OBSERVATIONS

1. Message traveled through authorized Google Workspace mail transfer agents in 23 seconds.
2. SPF authentication evaluated to PASS for sending IP 209.85.220.69 / 209.85.220.41.
3. DKIM cryptographic signature was signed by d=klu.ac.in and validated as PASS.
4. DMARC policy evaluated to PASS.
5. IP lookup for 209.85.220.69 verified ownership by Google LLC (AS15169).

FINDINGS

- **Authentication Alignment:** Complete alignment between the visible sender (hodcse@klu.ac.in), SPF permitted sender IP block, and DKIM cryptographic signature (d=klu.ac.in) verifies email authenticity.
- **Absence of Spoofing:** No anomalous routing hops, header modifications, or domain impersonation indicators were present.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The email header was extracted, parsed, and verified. SPF, DKIM, and DMARC checks confirmed message authenticity without spoofing.

CONCLUSION

Header analysis confirmed that the email originated from an authorized Google Workspace mail server for domain `klu.ac.in`. Cryptographic authentication mechanisms (SPF, DKIM, DMARC) verified header and content integrity.

REFERENCES

- RFC 5322 (Internet Message Format)
- RFC 7208 (Sender Policy Framework)
- RFC 6376 (DomainKeys Identified Mail)
- RFC 7489 (DMARC)
- Digital Forensics Laboratory Manual: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)